

Team 41

Student Grade Checker

CSE333 – Data Structures and Algorithms

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Project Idea & Objective

A GUI desktop application built with C++ and Qt that lets academic staff manage student records.

Core Features



Add Student



Remove Student



Search by ID



Update Grades



Sort by GPA



Top Student



Letter Grades



Input Validation

02 System Design & Architecture

main.cpp

QT ENTRY POINT

Initializes QApplication
and launches MainWindow

Student.h/.cpp

DATA MODEL

Stores ID, name, grades
Computes GPA + Letter Grade

GradeSystem.h/.cpp

BACKEND MANAGER

unordered_map storage
CRUD + search + sort logic

MainWindow

PRIMARY GUI

Table view, toolbar, status bar
Orchestrates all user actions

AddStudentDialog

ADD STUDENT POPUP

Collects ID, name, grades
Full inline validation

UpdateGradesDialog

UPDATE GRADES POPUP

Replaces student grades
Auto-recalculates GPA

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Data Structures Used & Why

```
GradeSystem::students
```

unordered_map
<int, Student>

O(1) average insert, search & delete by student ID — fastest for key-based lookup

```
Student::grades
```

vector<float>

Variable-length grades per student — size unknown at compile time, dynamic resizing

```
getAllStudents() / sortByGPA()
```

vector<Student>

Temporary collection for display & sorting — supports random access and std::sort

```
Student::name / toString()
```

std::string

Variable-length text storage for names and formatted output — simple, standard

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GUI

Main Window

The screenshot shows a window titled "Student Grade Checker" with standard Windows window controls (minimize, maximize, close). The window contains a search bar with the placeholder text "Search by Student ID...", a "Search" button, and a "Sort by GPA" button. Below these is a table with five columns: "ID", "Name", "Grades", "GPA", and "Grade". The table is currently empty. At the bottom of the window, there is a status bar that says "Total Students: 0" and a row of five buttons: "Add Student", "Update Grades", "Remove Student", "Top Student", and "Refresh".

ID	Name	Grades	GPA	Grade
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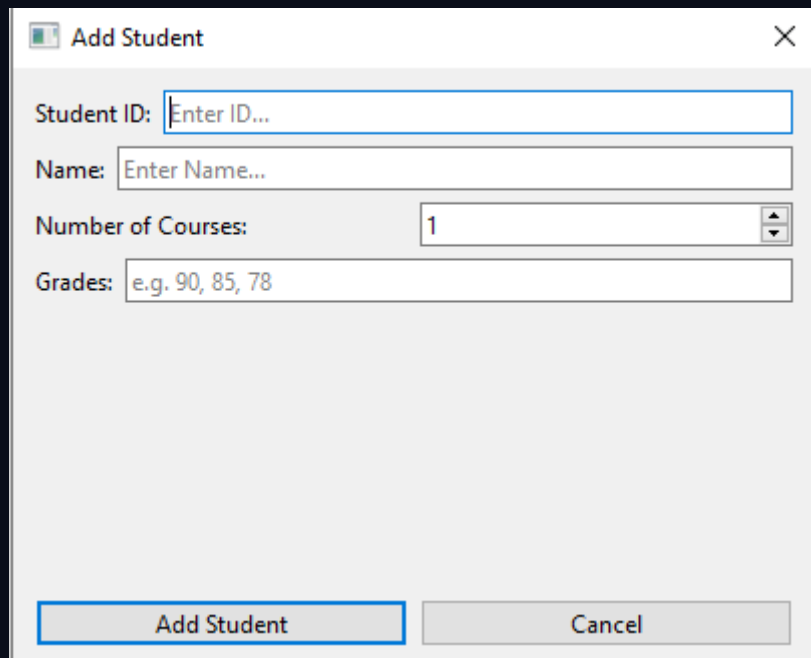
Total Students: 0

Add Student Update Grades Remove Student Top Student Refresh

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GUI

Add Student Dialog



A screenshot of a Windows-style dialog box titled "Add Student". The dialog has a title bar with a close button (X) in the top right corner. Inside the dialog, there are four input fields: "Student ID:" with a placeholder "Enter ID...", "Name:" with a placeholder "Enter Name...", "Number of Courses:" with a numeric spinner box containing the value "1", and "Grades:" with a placeholder "e.g. 90, 85, 78". At the bottom of the dialog, there are two buttons: "Add Student" and "Cancel".

Student ID: Enter ID...

Name: Enter Name...

Number of Courses: 1

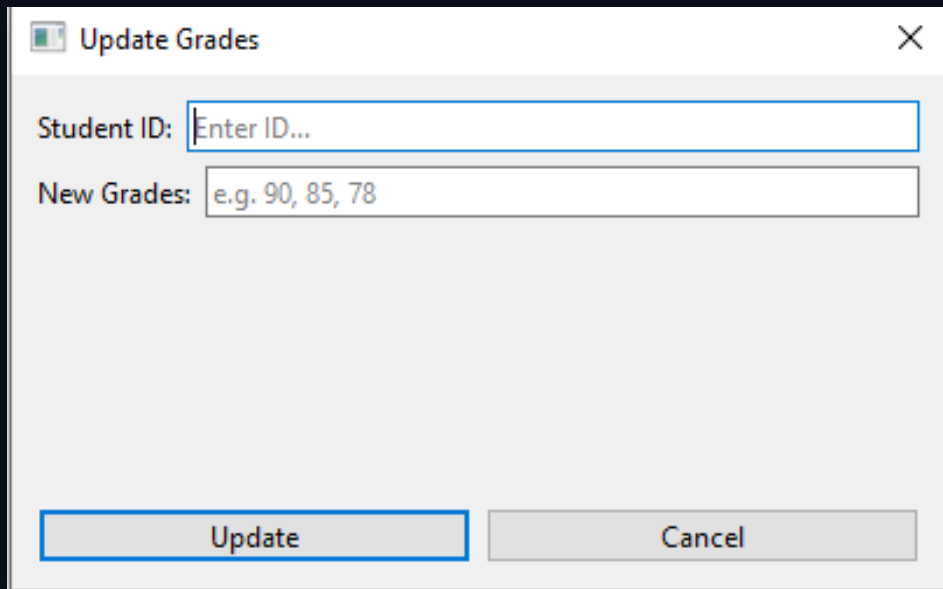
Grades: e.g. 90, 85, 78

Add Student Cancel

04

GUI

Update Grades Dialog



A screenshot of a Windows-style dialog box titled "Update Grades". The dialog has a title bar with a close button (X) in the top right corner. Inside the dialog, there are two text input fields. The first field is labeled "Student ID:" and contains the placeholder text "Enter ID...". The second field is labeled "New Grades:" and contains the placeholder text "e.g. 90, 85, 78". At the bottom of the dialog, there are two buttons: "Update" and "Cancel".

Update Grades

Student ID:

New Grades:

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AI Usage & Lessons Learned

AI WAS USED FOR

Feature Recommendations

sortByGPA, showTopStudent, searchStudent — group selected what fit scope

Test Plan Design

Happy path, edge cases, duplicate IDs, invalid input — group executed all manually

Documentation

README written with AI assistance based on group's own descriptions

Code Review

AI reviewed code; group evaluated every suggestion independently

SUGGESTIONS REJECTED

saveToFile() / loadFromFile()

Out of scope — kept focus on core features

priority_queue for Top Student

Understood the concept, kept $O(n)$ scan for readability

KEY TAKEAWAY

Every AI suggestion was evaluated, tested, and adopted, modified, or rejected.

"Using AI as an assistant does not replace understanding. AI output must be tested and verified."

Thank You

<https://github.com/lobnaahmed850/Student-Grade-Checker.git>